

Apéndice E. Hoja de datos del LM7805



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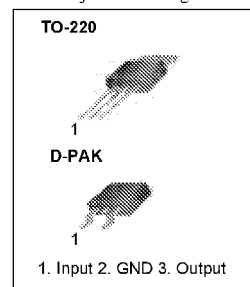
MC78XX/LM78XX/MC78XXA 3-Terminal 1A Positive Voltage Regulator

Features

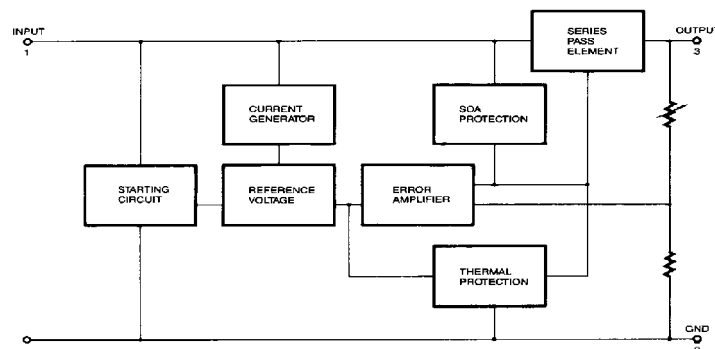
- Output Current up to 1A
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V
- Thermal Overload Protection
- Short Circuit Protection
- Output Transistor Safe Operating Area Protection

Description

The MC78XX/LM78XX/MC78XXA series of three terminal positive regulators are available in the TO-220/D-PAK package and with several fixed output voltages, making them useful in a wide range of applications. Each type employs internal current limiting, thermal shut down and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.



Internal Block Diagram



Rev. 1.0.1

MC78XX/LM78XX/MC78XXA

Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Input Voltage (for $V_O = 5V$ to $18V$) (for $V_O = 24V$)	V_I V_I	35 40	V V
Thermal Resistance Junction-Cases (TO-220)	$R_{\theta JC}$	5	$^{\circ}C/W$
Thermal Resistance Junction-Air (TO-220)	$R_{\theta JA}$	65	$^{\circ}C/W$
Operating Temperature Range	T_{OPR}	$0 \sim +125$	$^{\circ}C$
Storage Temperature Range	T_{STG}	$-65 \sim +150$	$^{\circ}C$

Electrical Characteristics (MC7805/LM7805)

(Refer to test circuit , $0^{\circ}C < T_J < 125^{\circ}C$, $I_O = 500mA$, $V_I = 10V$, $C_I = 0.33\mu F$, $C_O = 0.1\mu F$, unless otherwise specified)

Parameter	Symbol	Conditions	MC7805/LM7805			Unit
			Min.	Typ.	Max.	
Output Voltage	V_O	$T_J = +25^{\circ}C$	4.8	5.0	5.2	V
		$5.0mA \leq I_O \leq 1.0A$, $P_O \leq 15W$ $V_I = 7V$ to $20V$	4.75	5.0	5.25	
Line Regulation (Note1)	Regline	$T_J = +25^{\circ}C$				mV
		$V_O = 7V$ to $25V$ $V_I = 8V$ to $12V$	-	4.0 1.6	100 50	
Load Regulation (Note1)	Regload	$T_J = +25^{\circ}C$				mV
		$I_O = 5.0mA$ to $1.5A$ $I_O = 250mA$ to $750mA$	-	9 4	100 50	
Quiescent Current	I_Q	$T_J = +25^{\circ}C$	-	5.0	8.0	mA
Quiescent Current Change	ΔI_Q	$I_O = 5mA$ to $1.0A$	-	0.03	0.5	mA
		$V_I = 7V$ to $25V$	-	0.3	1.3	
Output Voltage Drift	$\Delta V_O / \Delta T$	$I_O = 5mA$	-	-0.8	-	mV/ $^{\circ}C$
Output Noise Voltage	V_N	$f = 10Hz$ to $100KHz$, $T_A = +25^{\circ}C$	-	42	-	$\mu V/V_O$
Ripple Rejection	RR	$f = 120Hz$ $V_O = 8V$ to $18V$	62	73	-	dB
Dropout Voltage	V_{Drop}	$I_O = 1A$, $T_J = +25^{\circ}C$	-	2	-	V
Output Resistance	r_O	$f = 1KHz$	-	15	-	$m\Omega$
Short Circuit Current	I_{SC}	$V_I = 35V$, $T_A = +25^{\circ}C$	-	230	-	mA
Peak Current	I_{PK}	$T_J = +25^{\circ}C$	-	2.2	-	A

Note:

1. Load and line regulation are specified at constant junction temperature. Changes in V_O due to heating effects must be taken into account separately. Pulse testing with low duty is used.

MC78XX/LM78XX/MC78XXA

Typical Performance Characteristics

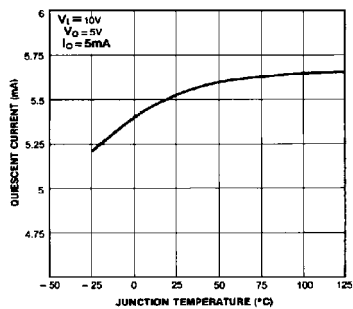


Figure 1. Quiescent Current

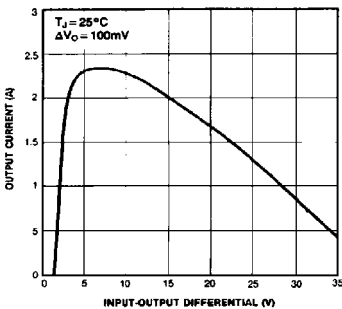


Figure 2. Peak Output Current

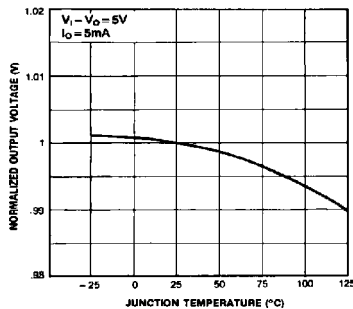


Figure 3. Output Voltage

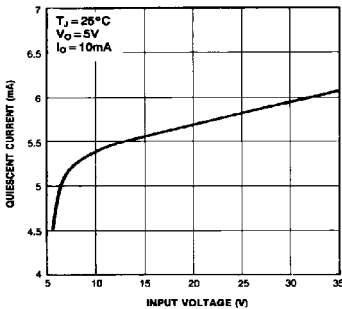


Figure 4. Quiescent Current

MC78XX/LM78XX/MC78XXA

Typical Applications

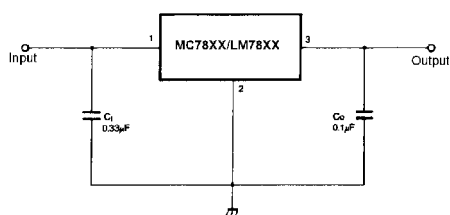


Figure 5. DC Parameters

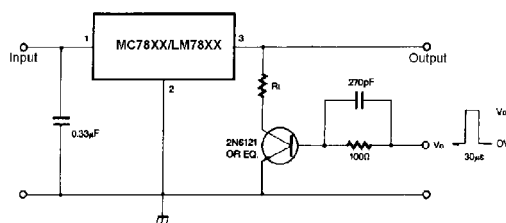


Figure 6. Load Regulation

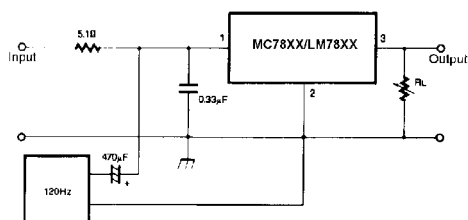


Figure 7. Ripple Rejection

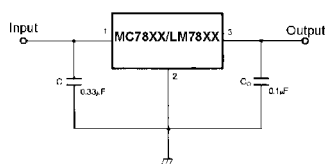


Figure 8. Fixed Output Regulator